School of Informatics

Informatics Project Proposal

Hybrid Topic and Domain-Adaptive Modelling for Financial Risk and Forecasting

UID: S2880814

Date: April 2026

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Abstract

Corporate risk disclosures contain dense, forward-looking information that general-purpose language models systematically misrepresent due to their insensitivity to domain-specific financial vocabulary. This project addresses that gap by developing a hybrid, domain-adapted sentence encoder trained on SEC Item~1A risk-factor filings. The encoder combines domain-adaptive pretraining with a novel three-view contrastive objective that extends prior dual-view formulations with an inter-firm sector signal, producing representations that capture regulatory, operational, and market risk semantics at efficient encoder scale. A BERTopic topic-modelling axis is integrated alongside the dense encoder to yield interpretable risk-cluster features, enabling the system to both predict and explain firm-level volatility. All adaptation and evaluation are conducted under a strictly forward-looking temporal protocol with training on 2006–2024 filings and evaluating on 2025–2026, to prevent temporal leakage from base-model pre-training data. A systematic multi-stage ablation isolates the contribution of each design choice and benchmarks the full hybrid system against SBERT and Fin-E5 under the standardised FinMTEB evaluation protocol.

Declaration of Own Work

I hereby declare that the work presented in this document is my own and the use of GenAI such as Claude/Grammarly is only carried out for spell checking and proofreading grammar.

1. Introduction

Corporate disclosures contain rich, forward-looking information about firm-level risks, yet these remain underexploited in quantitative finance. The dominant approaches such as bag-of-words sentiment lexicons [1] and general-purpose language models were not designed for the specialised vocabulary of risk disclosure words such as *liability*, *uncertainty*, and *exposure*. These vocabulary carry precise domain-specific meanings that general encoders systematically misrepresent. Recent work has begun to close this gap. Chiu et al. [2] demonstrate that contrastive adaptation on Item~1A (the mandatory risk-disclosure section of SEC annual filings) improves inter-firm risk retrieval. Jehnen et al. [3] fine-tune a sentence-transformer on Form~10-K (the SEC-mandated annual report) disclosures for corporate performance prediction, and Tang and Yang [4] introduce Fin-E5 (a financial-domain sentence embedding model) alongside FinMTEB (Financial Massive Text Embedding Benchmark) for standardised financial embedding evaluation. Yet no existing work combines risk-factor-specific adaptation at efficient encoder scale, a forward-looking temporal evaluation protocol, and a systematic multi-stage ablation. This project targets that remaining gap through a hybrid pipeline integrating domain-adaptive pretraining, contrastive fine-tuning, and BERTopic (a neural topic model) [5] over Item~1A filings.

2. State of the Art

This study builds a domain-adapted sentence embedding model for financial risk language and evaluates it on downstream volatility forecasting under strict forward-looking temporal splits. The primary data source is the Item~1A: Risk Factors section of SEC Form~10-K filings, accessed via EDGAR (Electronic Data Gathering, Analysis, and Retrieval, the SEC's public filing database) a forward-looking corporate disclosures covering regulatory, operational, credit, and market risks. General-purpose encoders such as SBERT (Sentence-BERT; [6]) and E5 (a large-scale dense sentence encoder trained with contrastive objectives [7]) are insensitive to financial risk vocabulary, while token-level financial models such as FinBERT (a BERT model fine-tuned on financial text for sentiment classification) target sentiment rather than risk-level semantic similarity. Fin-E5 [4] advances the state of the art in financial sentence embeddings but relies on a 7B-parameter large language model (LLM) backbone, making it impractical for deployment at scale, and is trained on broad financial text rather than risk-factor language specifically. Closest to this project, Chiu et al. [2] apply dual-view contrastive adaptation to Item~1A filings, and Jehnen et al. [3] fine-tune a sentence-transformer on 10-K disclosures for corporate performance prediction, yet neither enforces a forward-looking temporal split nor ablates across adaptation stages. Table 1 summarises the nine works surveyed above. The residual gap is therefore precise: an Item~1A specific encoder, trained at efficient scale, evaluated under strictly forward-looking conditions (training capped at 2024; evaluation on 2025–2026 filings to prevent temporal leakage from base-model pre-training data), with ablation isolating the contribution of each adaptation stage and a BERTopic (neural topic model) [5] axis providing interpretable risk structure alongside forecasting signals.

Table 1: Summary of literature survey

Reference & Venue	**Dataset Used**	**Type of Representation**	**Adaptation Method**	**Downstream Task**	**Evaluation Protocol**
Loughran and McDonald (2011) [1] *Journal of Finance*, 66(1)	SEC 10-K filings (EDGAR, 1994–2008; 50,115 documents)	Bag-of-words; finance-specific sentiment word lists (TF-IDF weighted)	None; hand-crafted domain wordlists calibrated to financial vocabulary	Return prediction; trading volume; volatility regression; fraud detection	Filing-date abnormal returns; R^2 in OLS regressions; word frequency statistics
Gururangan et al. (2020) [8] *ACL 2020*	Biomedical, CS, news, and review corpora (domain-specific unlabelled text)	Contextual token embeddings (RoBERTa); task-specific classifier head	DAPT: continued MLM on domain corpus; TAPT: continued MLM on task corpus	Text classification across 8 tasks (4 domains)	Macro-F1; comparison of DAPT, TAPT, and DAPT+TAPT conditions
Reimers and Gurevych (2019) [6] *EMNLP 2019*	SNLI, MultiNLI (general domain); STS-B for evaluation	Fixed-dim sentence embeddings via siamese/triplet BERT networks	Fine-tuning on NLI classification and STS regression with siamese architecture	Semantic textual similarity; information retrieval; clustering	Spearman ρ on STS-B; inference time (sec per n pairs)
Wang et al. (2022) [7] – E5 @ *ICLR 2024*	Billions of weakly-supervised (query, passage) pairs from web corpora (MS-MARCO, CCNet)	Dense sentence embeddings (encoder + contrastive loss); 1,024-dim vectors	Weakly-supervised contrastive pre-training at scale; supervised fine-tuning	Retrieval, reranking, classification, STS	BEIR benchmark (NDCG@10, Recall@100); MTEB (Massive Text Embedding Benchmark) average
Grootendorst (2022) [5] arXiv†	20 Newsgroups; general corpora	Document embeddings (BERT/SBERT) + c-TF-IDF (class-based TF-IDF) cluster term weights	None (modular; any encoder pluggable as backend)	Topic modelling; document clustering	Topic coherence (C_V); topic diversity; qualitative inspection
Chiu et al. (2025) [2] *EMNLP 2025*	SEC Form 10-K filings (Item 1A: Risk Factors); multiple firms and years	Contrastive sentence encoder (Jina AI base); dense financial risk vectors	Unsupervised dual-view contrastive: lexical view (overlapping spans) + chronological view (shared date tokens, masked)	Inter-firm risk relation identification; portfolio signals	Retrieval precision; Risk Relation Score (RRS) stability across thresholds

Reference & Venue	Dataset Used	Type of Representation	Adaptation Method	Downstream Task	Evaluation Protocol
Tang and Yang (2025) [4] FinMTEB + Fin-E5 @ *EMNLP 2025*	64 financial datasets: regulatory filings, news, ESG reports, earnings transcripts	LLM-as-encoder (e5-mistral-7b backbone); dense sentence vectors; 4,096-dim	Persona-based synthetic augmentation + multi-stage contrastive fine-tuning	STS, retrieval, classification, clustering, reranking, summarisation	FinMTEB average (MAP, NDCG@10, F1, Spearman~ρ) across 7 tasks
Yang, Liu and Wang (2023) [9] *FinLLM @ IJCAI 2023*	Real-time financial data: news wires, social media, earnings calls, SEC filings (multi-source; continuously updated pipeline)	Generative LLM (LLaMA backbone) with parameter-efficient LoRA (Low-Rank Adaptation) adapters; full end-to-end open-source framework	LoRA-based RLHF (Reinforcement Learning from Human Feedback) fine-tuning on curated financial corpora; automated data acquisition, cleaning, and preprocessing pipeline	Sentiment analysis; robo-advising; algorithmic trading; low-code financial analysis	Task-specific benchmarks; qualitative and quantitative comparison vs. BloombergGPT and GPT-4 on financial NLP tasks
Jehnen, Villalba-Diez and Ordieres-Meré (2026) [3] *Frontiers in AI*	S&P~500 10-K filings (Item~7 and Item~7A; 2016–2023)	Domain-specific sentence-transformer (fine-tuned encoder) paired with BERTopic for latent topic extraction	Full fine-tuning of a sentence-transformer on financial disclosure text; benchmarked against LDA, NMF, and off-the-shelf encoders	Topic modelling of financial disclosures; corporate performance prediction via logistic regression / MLP	Intratopic / intertopic similarity; ROC-AUC and F1-score; 2 pp improvement over general-purpose baseline

†BERTopic is an arXiv preprint only. Retained due to its status for neural topic modelling (>4,000 peer-reviewed citations).

3. Implementation

3.1. Methodology or Approach

The methodology comprises five phases, extending the dual-view contrastive framework of Chiu et al. [2] with an upstream domain-adaptive pretraining (DAPT) stage [8] and a downstream BERTopic topic-modelling axis [5]. Each design choice is grounded in prior evidence: DAPT addresses documented vocabulary mismatch [1, 8], three-view contrastive fine-tuning extends a proven unsupervised framework [2] with an additional inter-firm signal, BERTopic provides interpretable risk structure that classical topic models cannot match at this scale [3], and the strict forward-looking temporal split prevents the base-model leakage that undermines most existing evaluations. Together, these choices directly address the three-part gap identified in Section 2.

3.1.1 Data pipeline

Item 1A is selected as the sole text source because it is the legally-mandated, forward-looking risk-disclosure section of Form 10-K [1]. Sections are scraped from EDGAR, sentence-tokenised, and partitioned by a strict temporal split (2006–2024 train / 2025 val / 2026 test). Training is capped at 2024 because base language model pre-training data typically extends through this period and evaluating on 2024 outcomes therefore risks implicit contamination from knowledge encoded during base pre-training. Restricting evaluation to 2025–2026 filings ensures downstream targets are strictly unseen by any component of the pipeline. The 30-day forward realised volatility window is adopted as the prediction target because it is the standard post-filing market-reaction window in empirical finance which captures the price uncertainty attributable to disclosed risks before earnings-cycle noise dominates. Three firm-level linkages are constructed:

1. **CIK–PERMNO** for volatility label assignment (CIK: Central Index Key, the SEC's unique firm identifier and PERMNO: CRSP's permanent stock number)
2. **CIK–SIC** via the EDGAR company submissions API, required for sector-view positive-pair construction in Phase 3 (SIC: Standard Industrial Classification, 2-digit)
3. a **fiscal-year index** aligning filing dates to return windows

3.1.2 Domain-adaptive pretraining (DAPT)

Although `all-mpnet-base-v2` is the strongest general-purpose sentence encoder at base (110M parameter) scale, its pretraining corpus systematically underrepresents financial risk vocabulary [1]. Gururangan et al. [8] demonstrate that continued masked-language modelling (MLM) on domain text consistently improves downstream task performance, even for models already trained at scale. DAPT is therefore applied first to reshape the token representations before any supervised signal is introduced. `all-mpnet-base-v2` specifically is chosen as the base because it serves as the SBERT (base) substitute in the ablation. Keeping the architecture constant across all conditions means only the adaptation strategy varies, producing a cleaer comparison. MLM is run at the standard 15% masking rate and validation perplexity tracked via Weights & Biases (W&B) guides checkpoint selection.

3.1.3 Contrastive fine-tuning

DAPT reshapes token representations but does not optimise sentence-level embedding geometry. Contrastive fine-tuning directly address this by pulling together semantically similar sentences and pushing apart dissimilar ones in the embedding space [6], which is the structure required for downstream retrieval and similarity tasks. Building on Chiu et al.'s [2] dual-view framework, this project introduces a third *sector view.* Chiu et al.'s lexical and chronological views operate entirely within a single filing and can encode intra-document risk phrasing and temporal persistence, but they cannot represent inter-firm risk structure. Yet firms within the same industry share regulatory environments, supply-chain dependencies, and macroeconomic exposures. Hence, their Item~1A disclosures therefore exhibit systematic vocabulary overlap that transcends any individual document [1].

Treating sentences from firms sharing the same 2-digit SIC code within the same fiscal year as soft positives and cross-sector pairs as hard negatives, encodes this inter-firm sectoral co-movement directly into the embedding space. The resulting three-view signal (lexical overlap, chronological view, sectoral co-movement) is a strictly richer contrastive objective than the original two-view formulation. SimCSE (Simple Contrastive Sentence Embeddings) in-batch negatives provide the contrastive loss, as they avoid expensive offline mining while yielding high-quality hard negatives from the batch [2].

In parallel, a LoRA-adapted (Low-Rank Adaptation) [9] variant is trained using the same three-view objectives but with rank-decomposed adapter layers in place of full parameter updates. Yang et al. [9] show this achieves competitive performance under compute constraints, motivating it as a parameter-efficient ablation condition in Phase 5. Intrinsic evaluation uses FinMTEB STS and retrieval sub-tasks [4], enabling direct comparison with Fin-E5 and SBERT under a standardised protocol.

3.1.4 BERTopic topic extraction

Classical topic models such as LDA (Latent Dirichlet Allocation) and NMF (Non-negative Matrix Factorisation) rely solely on word co-occurrence statistics and cannot capture the semantic similarity structure that distinguishes. BERTopic [5] replaces co-occurrence with neural document embeddings then applies c-TF-IDF (class-based TF-IDF) to label each cluster with its most distinctive terms thus producing interpretable topic names that reflect financial risk meaning rather than surface word frequency [3]. Crucially, using the *domain-adapted* encoder (rather than generic SBERT) as the BERTopic backend means clusters reflect financial risk similarity rather than general English similarity. This phase is run before downstream ablation so that topic-cluster features are available as a second input representation in the ablation that follows. Topic coherence (C_V) and diversity are reported against a generic-SBERT-backend baseline to quantify the added value of domain adaptation on topic quality.

3.1.5 Downstream evaluation and ablation

With both dense embeddings and topic-cluster features in hand, the full ablation is run. Mean-pooled filing embeddings (or topic-cluster feature vectors, or both) are fed to a ridge regression head and a multi-layer perceptron (MLP) head predicting 30-day forward volatility. Ridge regression is included as a linear probe because it tests whether the embedding signal is already linearly decodable for volatility. If it is, the representations carry sufficient structure for a downstream practitioner to exploit without a deep predictor. The MLP tests whether non-linear interactions provide additional predictive power, ensuring the ablation conclusions are robust to head choice.

Seven conditions are evaluated: (1) SBERT (no adaptation, lower-bound anchor), (2) Fin-E5 (best existing financial encoder, upper-bound anchor), (3) DAPT-only (does MLM on risk text help?), (4) DAPT + contrastive (does contrastive fine-tuning add further?), (5) LoRA-adapted (does parameter-efficient adaptation match full fine-tuning?), (6) topic-features-only (do BERTopic clusters alone predict volatility?), and (7) hybrid (embeddings + topics, testing the core project claim).

Spearman ρ is the primary metric because rank correlation is robust to the heavy-tailed distribution of realised volatility; MSE, R^2, and Sharpe ratio provide complementary absolute and finance-practitioner-oriented measures.

3.2. Expected Outcomes

This project is expected to produce five primary deliverables, each corresponding directly to a project phase. At the model level, the domain-adapted encoder (`all-mpnet-base -v2` after DAPT and three-view contrastive fine-tuning) is expected to outperform the unadapted SBERT baseline on FinMTEB [4] STS and retrieval sub-tasks, and to match or exceed Fin-E5 under the same standardised protocol (a significant result given that Fin-E5 uses a 7B-parameter LLM backbone versus the 110M-parameter encoder used here). At the topic level, BERTopic [5] applied to Item 1A filings via the domain-adapted encoder is expected to yield coherent, financially interpretable risk clusters (e.g. *regulatory litigation risk*, *supply-chain disruption*) with topic coherence (C_V) exceeding the general-purpose SBERT baseline. At the forecasting level, the hybrid condition (dense embeddings + topic-cluster features) is expected to achieve a higher Spearman ρ on 30-day forward volatility prediction than either representation alone. Finally, the ablation study is expected to produce a comparative table isolating the marginal contribution of each adaptation stage (DAPT, contrastive fine-tuning, LoRA, BERTopic axis), directly resolving the methodological gap identified in the State of the Art.

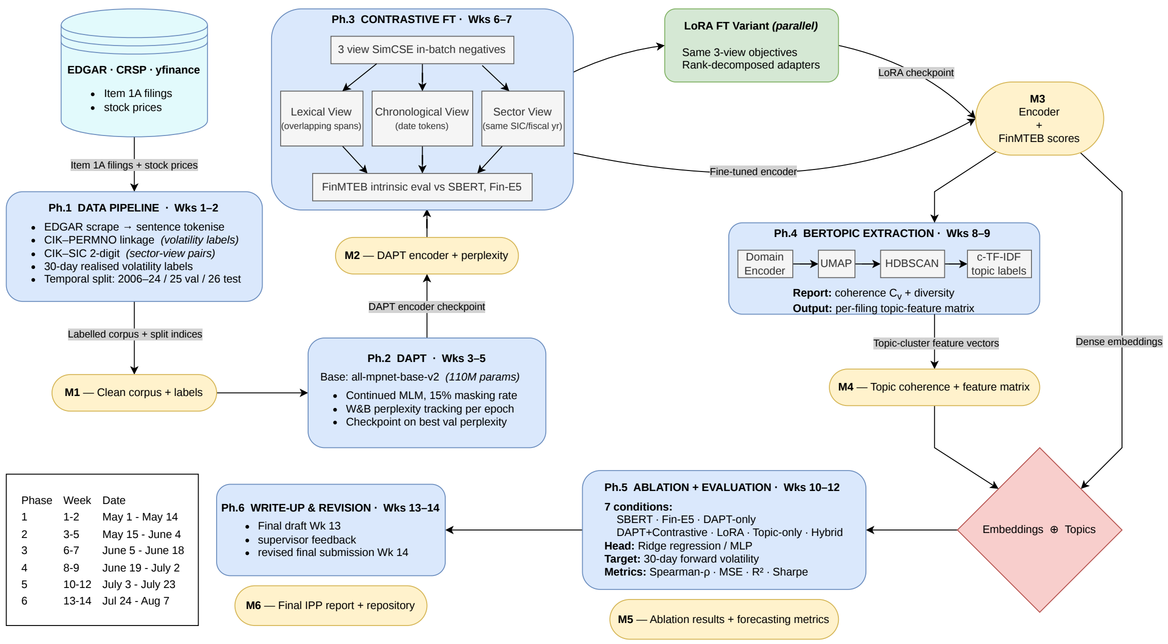

Figure 1: Five-phase methodology pipeline for hybrid financial risk modelling

3.3. Evaluation Criteria

Table 2 summarises the success criteria for each project phase, specifying the metric, the comparison baseline, and the measurement protocol.

Table 2: Evaluation criteria by project phase.

Phase	Metric	Baseline	Success Criterion	Measurement Protocol
2. DAPT	Validation perplexity (MLM)	`all-mpnet-base-v2` zero-shot perplexity on held-out risk sentences	Monotonically decreasing across epochs; final perplexity $<$ pre-DAPT value	W&B per-epoch logging; best checkpoint selected on val perplexity
3. Contrastive FT	Spearman ρ (STS); NDCG@10 (retrieval)	SBERT (`all-mpnet-base-v2`); Fin-E5	STS ρ and NDCG@10 exceed SBERT; competitive with or surpassing Fin-E5	FinMTEB STS + retrieval sub-tasks [4]
4. BERTopic	Topic coherence C_V; topic diversity	BERTopic with unadapted SBERT encoder	$C_V > 0.5$; diversity > 0.7; topics labelled as financially meaningful on qualitative inspection	BERTopic built-in coherence/diversity scorer; manual label review
5. Ablation	Spearman ρ; MSE; R^2; Sharpe ratio	SBERT (no adaptation); Fin-E5 (best existing)	Hybrid condition Spearman $\rho >$ any single-representation condition; $R^2 > 0$ on 2025–2026 test set	Ridge regression / MLP head; held-out 2025–2026 filings; per-condition result table

3.4. Responsible Research Including Ethics

All data are sourced from publicly available, regulatory-mandated SEC filings (EDGAR) and aggregated market price feeds (CRSP/`yfinance`). No personal data, human subjects, or proprietary datasets are used, and no informed consent or IRB review is required. CRSP data will be accessed exclusively through WRDS under the University's institutional licence and will not be redistributed or retained beyond the project period.

3.5. Risk Assessment and Mitigations

Table 3 identifies the principal project risks, assessed after mitigations are in place following the likelihood × severity matrix: High × High = Extreme; High × Medium or Medium × High = High; Medium × Medium = Medium; Low × any = Low/Very Low.

Table 3: Project risk assessment (post-mitigation ratings).

Activity	**Risk**	**Affects**	**Risk Likelihood**	**Risk Severity**	**Control Measures**	**Overall Risk Category**
Phase 1: Data Access (WRDS/CRSP)	WRDS account not approved before pipeline build begins; CRSP volatility labels unavailable	Researcher; timeline	Medium	Medium	EDGAR scraping (free) begins immediately; `yfinance` Python package provides stock-price fallback for S&P 500 firms; CIK–SIC via EDGAR submissions API (no WRDS dependency); WRDS access requested awaiting response	Medium
Phase 2: Teaching Cluster (DAPT)	GPU job queue congestion or allocation unavailable during 3-week DAPT window; job eviction before convergence	Researcher; timeline	High	Medium	Submit jobs early in each week; checkpoint saved every epoch so evicted jobs resume cleanly; fallback: Google Colab Pro (T4/A100 free tier) for smaller batch runs	High
Phase 3: Contrastive Fine-tuning	Custom three-view sector-pair construction contains bugs (incorrect SIC groupings, label leakage across train/val)	Project outcomes	Medium	High	Unit-test pair construction on toy corpus before full training; validate SIC groupings against EDGAR API ground truth; inspect positive/negative pair distributions before each run	High

Activity	Risk	Affects	Risk Likelihood	Risk Severity	Control Measures	Overall Risk Category
Phase 3: LoRA Implementation	LoRA adapter configuration (rank, target modules) not tuned; underperforms full fine-tuning baseline	Ablation validity	Low	Low	Follow FinGPT [9] recommended ranks ($r \in \{8, 16\}$); treat LoRA as an exploratory ablation condition, not a primary claim	Very Low
Phase 4: BERTopic Compute	UMAP dimensionality reduction on large embedding matrix exceeds Teaching Cluster RAM (~32 GB node limit)	Researcher; timeline	Low	Low	Subsample corpus for UMAP fit (10k–20k documents); use `cuml` GPU UMAP if available; reduce `n_components` (5–10)	Very Low
Phase 5: 2026 Test-Set Coverage	FY2025 10-K filings filed January–April 2026; many firms will not have filed by project start, yielding a smaller test set than planned	Evaluation validity	High	Low	Accept reduced test size; report confidence intervals on Spearman ρ; supplement with 2025 val set for sensitivity analysis; disclose limitation explicitly in write-up	Medium
Phases 1–4: Timeline Slippage	Earlier phases overrun, leaving insufficient time for Phase 5 ablation or write-up	Researcher; degree outcome	Medium	High	Phase 5 includes an explicit contingency buffer (Wks 10–12); MVP fallback: revert to two-view Chiu et al. framework, drop sector view and LoRA, reducing ablation to 5 conditions	High

4. Research Plan, Milestones, and Deliverables

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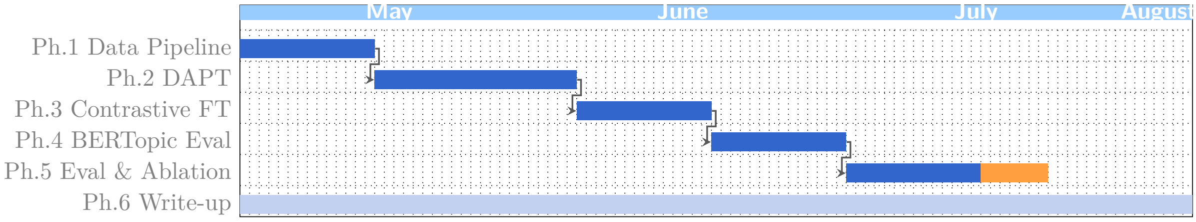

Figure 2: Gantt chart of project activities. ■ scheduled phase; ■ contingency buffer (Ph.5 only); ■ continuous write-up (parallel throughout). Arrows indicate sequential task dependencies.

Milestone	**Week**	**Description**
M_1	2	Clean corpus, temporal split indices, volatility labels
M_2	5	Domain-adapted encoder + perplexity benchmarks
M_3	7	Fine-tuned encoder + FinMTEB intrinsic scores
M_4	9	BERTopic topic coherence scores + per-filing topic-feature matrix
M_5	12	Full ablation results table + downstream forecasting metrics
M_6	14	Final IPP report + repository submission

Table 4: Project milestones.

Deliverable	**Week**	**Description**
D_1	5	Domain-adapted encoder (DAPT checkpoint)
D_2	7	Contrastive fine-tuned encoder + FinMTEB evaluation
D_3	9	BERTopic risk topic analysis + per-filing topic-feature matrix
D_4	12	Full ablation study + downstream volatility forecasting results
D_5	14	Final IPP report + reproducible codebase

Table 5: Project deliverables.

References

[1] Tim Loughran and Bill McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of finance*, 66(1):35–65, 2011. DOI: `https://doi.org/10.1111/j.1540-6261.2010.01625.x`.

[2] Wei-Ning Chiu, Yu-Hsiang Wang, Andy Hsiao, Yu-Shiang Huang, and Chuan-Ju Wang. Financial risk relation identification through dual-view adaptation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 26301–26311, 2025. DOI: `https://doi.org/10.18653/v1/2025.emnlp-main.1336`.

[3] Simon Jehnen, Javier Villalba-Díez, and Joaquín Ordieres-Meré. Fintextsim: a domain-specific sentence-transformer for extracting predictive latent topics from financial disclosures. *Frontiers in Artificial Intelligence*, 9:1752103, 2026. DOI: `https://doi.org/10.3389/frai.2026.1752103`.

[4] Yixuan Tang and Yi Yang. FinMTEB: Finance massive text embedding benchmark. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 3620–3638, Suzhou, China, November 2025. Association for Computational Linguistics. DOI: `https://doi.org/10.18653/v1/2025.emnlp-main.179`.

[5] Maarten Grootendorst. Bertopic: Neural topic modeling with a class-based tf-idf procedure. *arXiv preprint arXiv:2203.05794*, 2022. DOI: `https://arxiv.org/abs/2203.05794`.

[6] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019. DOI: `https://doi.org/10.18653/v1/D19-1410`.

[7] Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022. DOI: `https://arxiv.org/abs/2212.03533`.

[8] Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A Smith. Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 8342–8360, 2020. DOI: `https://doi.org/10.18653/v1/2020.acl-main.740`.

[9] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. FinGPT: Open-source financial large language models. FinLLM at IJCAI 2023, June 2023. DOI: `https://doi.org/10.2139/ssrn.4489826`.